# EEG1001 AGV project
# AGV Electronics Introduction

# AGV Electronics Overview

- What do you fabricate?
  - *Automatic* Guided Vehicle (AGV)
  - Needs onboard computer or *microcontroller* for decision making and control
- What is to be controlled?
  - Speed
  - Steering
  - User interaction
  - Navigation

# AGV Electronics Overview

- Drive motor control?
  - ESC: Electronic Speed Controller (really "torque" controller…)
  - *Provided* as integrated module for your project

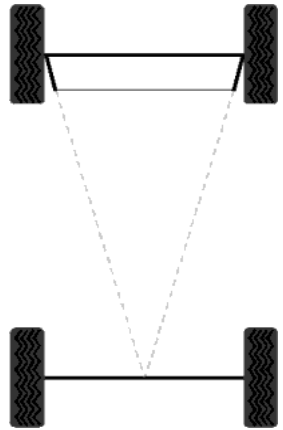



- How "powerful" is motor to be?
  - Can you estimate theoretically? (vehicle mass, acceleration, velocity...)

# AGV Electronics Overview

- Steering Control
  - Two-wheel steering: Ackerman condition
    - Mechanism (approximate): trapezoidal linkage
  - Control actuator:
    - Servomotor:

    Driven directly by onboard computer/microcontroller





# AGV Electronics Overview

- On-board *energy* source?
  - Battery is “obvious” choice?
  - Provided battery pack, nominal ratings: 7.2V, 2700 mAh, rechargeable (2700mAh=2.7Ah)
  - mAh (milli ampere hour) is the unit of charge

❖ If vehicle draws ~1 A, how long could it run on one charge?

# AGV Electronics Overview

- What is the Battery Energy Capacity?

$$Energy = Power \times Time$$

$$Power\ (P) = Voltage\ (V) \times Current\ (I)$$

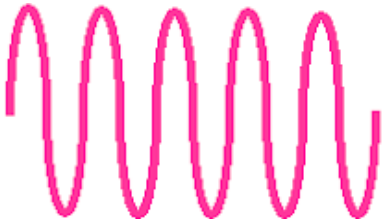
- If we draw 1A at 7.2 V, $P = 7.2 \times 1 = 7.2\ W$

- So the energy , $E = P \times time = 7.2 \times 1\ hr = 7.2\text{Wh}$

For 2.7 hrs, $E = 7.2 \times 2.7\ hr = ??$

# AGV Electronics Overview

- What on-board *computer* can we use?
  - Standard "computer" is too big (and high cost…)
  - Use a single-chip "microcontroller"
  - General purpose computer core plus integrated I/O support:
    - Digital I/O
    - Analog to Digital Conversion (ADC)
    - Digital to Analog Conversion (DAC)



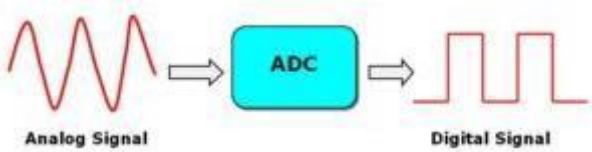
Analog Signal



Digital Signal

# AGV Electronics Overview

- ADC allows input from continuous valued sensors: temperature, pressure, speed…




- DAC allows continuous valued output: audio, steering, heating, illumination…




- *Provided* microcontroller(s):
  - Arduino Nano 33 BLE (Microcontroller module): cost= €22.8
  - Specifically: nRF52840

# AGV Electronics Overview

- *Potential* functionality constrained by *hardware*
  - processing speed, memory capacity, I/O capacity
  - Motion:
    - Steering and speed control
    - Optical sensor module (detect floor track)
  - User interaction
    - Light Emitting Diodes (LED): status indicators
    - Keypad and display
  - Fuse: to avoid specific hazards: fire, component damage
  - Interconnection/wiring harness

# AGV Electronics Overview

- *Actual* functionality defined by *software*
  - OpenMV IDE (Integrated development Environment)
  - Code editing, compiling, debugging, compiling and download to MCU

# Speed and Steering Control

- Both speed and steering should (ideally) be managed by “closed loop feedback control”
- Implemented in software running on the onboard computer
  - Maximum flexibility for adjustment/revision
- Will discuss more in semester 2